29) Construct a 3x3 matrix, not in echelon form, whose columns span $\mathbb{R}^3$. Show that the matrix you construct has the desired property.

$\boxed{\text{Claim:}}$ The matrix $A = \begin{bmatrix} 1 & 5 & 2 \\ 4 & 2 & 8 \\ 5 & 7 & 6 \end{bmatrix}$ is not in echelon form and its columns span $\mathbb{R}^3$.

$\boxed{\text{Proof:}}$ Row 1 Leading Entry = 1 in Column 1
Row 2 Leading Entry = 4 in Column 1
Row 3 Leading Entry = 5 in Column 1

Since in an echelon form of a matrix, the leading element of each row must be in a column to the right of the columns of the leading elements of all rows above it, the matrix A is not in echelon form.

From Theorem 4, the columns of A span $\mathbb{R}^3$ if and only if each row of A has a pivot position. By defn, a pivot position of a matrix corresponds to a leading 1 of that row in reduced row echelon form.

$$\begin{bmatrix} 1 & 5 & 2 \\ 4 & 2 & 8 \\ 5 & 7 & 6 \end{bmatrix} \xrightarrow[-4\times Row1]{Row2 = Row2} \begin{bmatrix} 1 & 5 & 2 \\ 0 & -18 & 0 \\ 5 & 7 & 6 \end{bmatrix} \xrightarrow[-5\times Row1]{Row3 = Row3} \begin{bmatrix} 1 & 5 & 2 \\ 0 & -18 & 0 \\ 0 & -18 & -6 \end{bmatrix} \xrightarrow[-Row2]{Row3 = Row3}$$

$$\begin{bmatrix} 1 & 5 & 2 \\ 0 & -18 & 0 \\ 0 & 0 & -6 \end{bmatrix} \xrightarrow[\times(-1/6)]{Row3 = Row3} \begin{bmatrix} 1 & 5 & 2 \\ 0 & -18 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow[-2\times Row2]{Row1 = Row1} \begin{bmatrix} 1 & 5 & 0 \\ 0 & -18 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow[\times(-1/18)]{Row2 = Row2}$$

$$\begin{bmatrix} 1 & 5 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow[-5\times Row2]{Row1 = Row1} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Verification of whether the final matrix is in reduced row echelon form

1. There are no zero rows in the matrix. $\therefore$ The condition that all non-zero rows must be above the zero rows is vacuously True.
2. Row 1 Leading Element = 1 in Column 1
Row 2 Leading Element = 1 in Column 2
Row 3 Leading Element = 1 in Column 3.